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Unit 3 - Hello, My Robot Friend

PDF Documents

PDF Document 1: 1739430769112-Chapter 2 - unit 3 - Hello, My Robot Friend.pdf

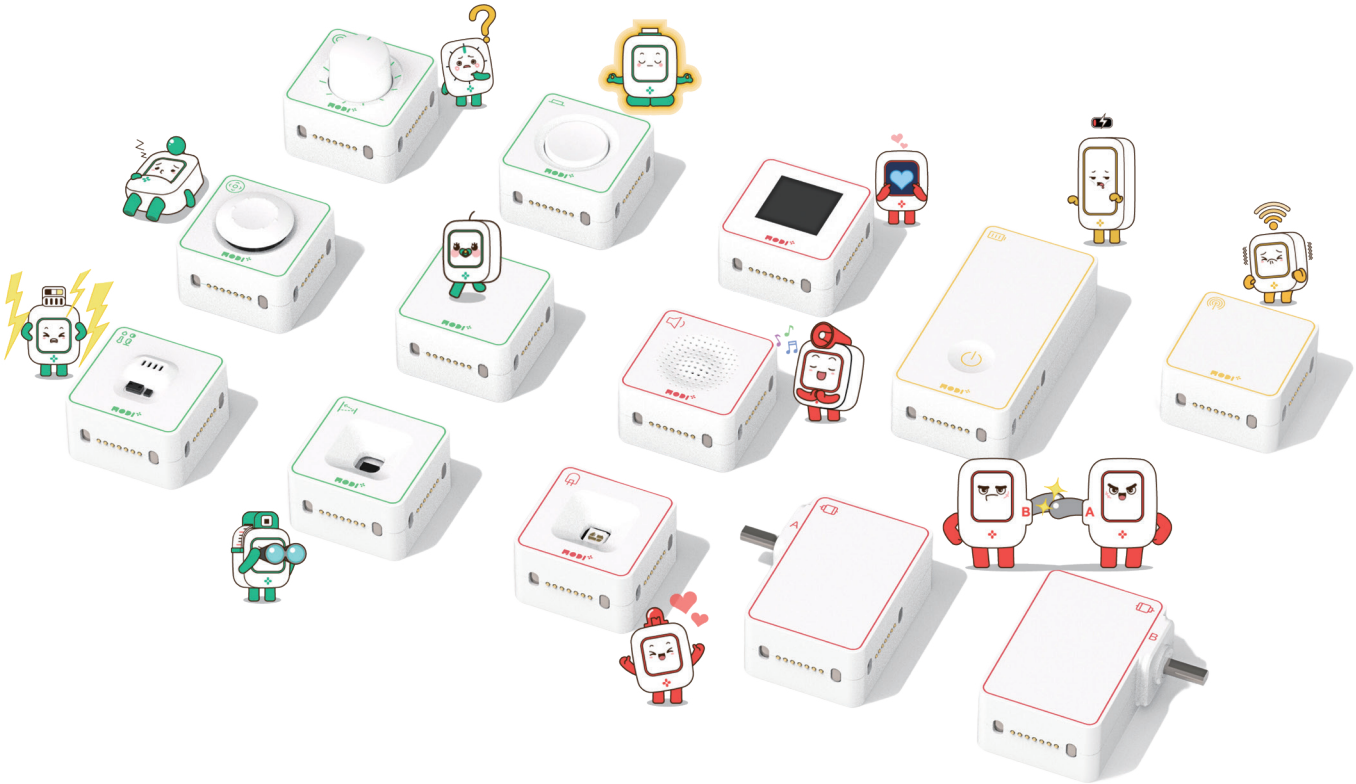
This PDF document is included in the unit materials.

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UNIT 3 HELLO, MY ROBOT FRIEND

Meet MODI: Our Robot Friend



Now, let's meet a particular robot named MODI. MODI is a smart robot that can do many different things. MODI can follow instructions and rules, just like the robots we talked about before.

How MODI Uses Inputs & Outputs & Setups

MODI can understand the world around it and interact with it using inputs and outputs.



1. INPUTS : Inputs are ways MODI gets information.

INPUT MODULES



Button

Senses the status of the button



Dial

Senses the rotation of the dial



Environment

Senses temperature, humidity, illuminance, and sound



IMU

Senses slopes (Angles)



ToF

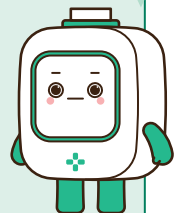
Senses distance



Joystick

Senses the direction of the stick

Inputs are green!



2. OUTPUTS : Outputs are ways MODI can respond or do something.

OUTPUT MODULES



Speaker

Make sounds



Motor A, B

Rotates



Display

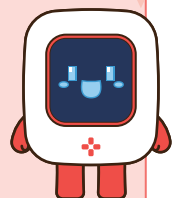
Shows letters and images



LED

Emits light

Outputs are red!



3. SETUPS : Setups are modules essential for operation

Setups are yellow!

SETUP MODULES



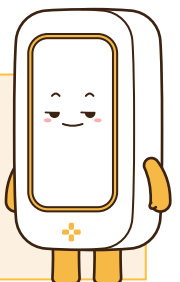
Battery

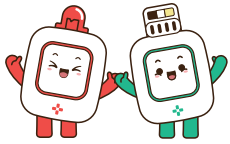
Supplies energy to the modules



Network

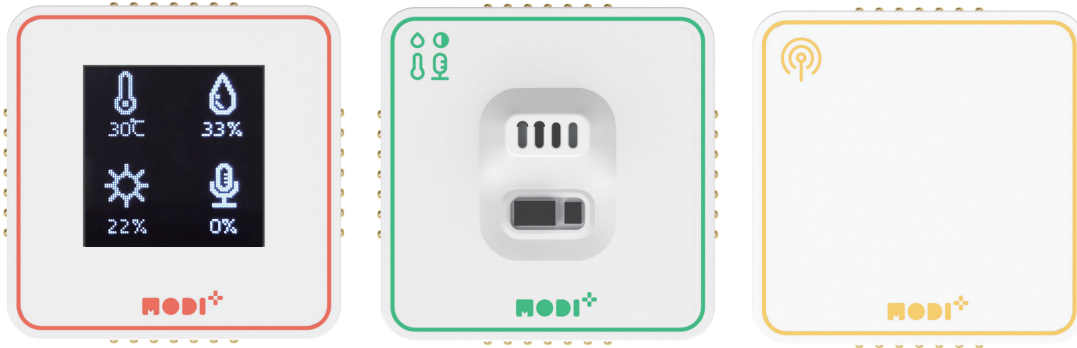
Connects modules to other modules, PCs, tablets, and phones





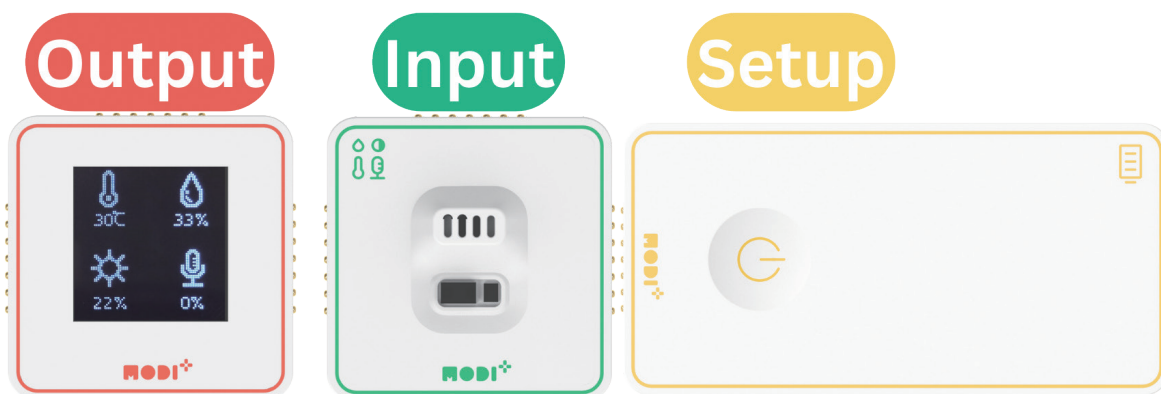
UNPLUGGED

Activity 1 Environment Meter



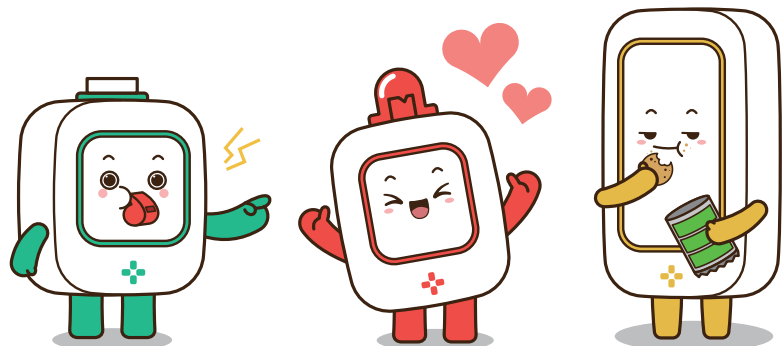
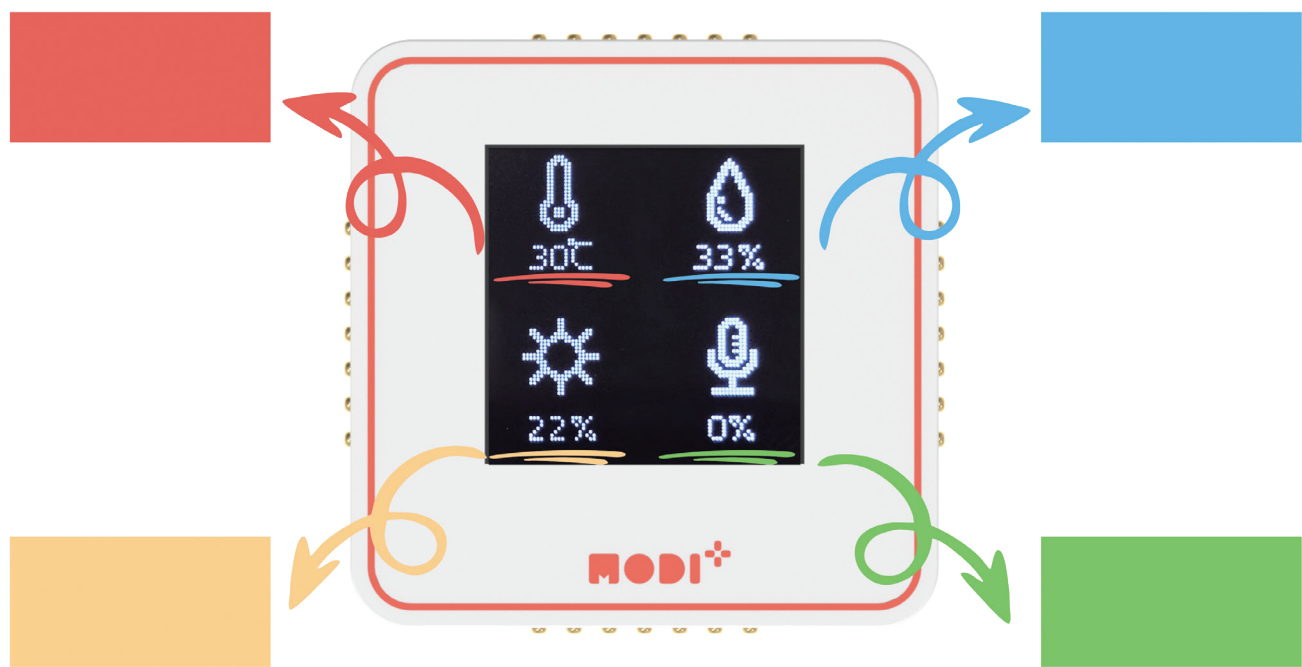
Instructions:

- 1) Connecting Your MODI Modules: Just bring your modules close together, and they'll snap into place like magic, thanks to the magnets on their sides.
- 2) Powering Up Your Modules: There are two cool ways to make your modules come alive:
Use the Battery module for power for the connected modules.





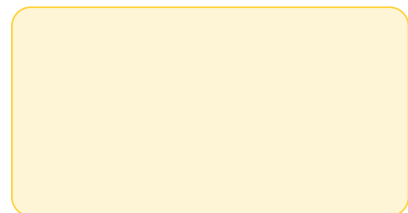
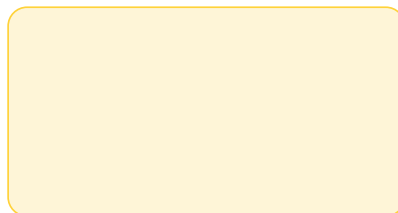
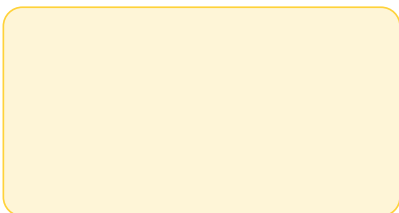
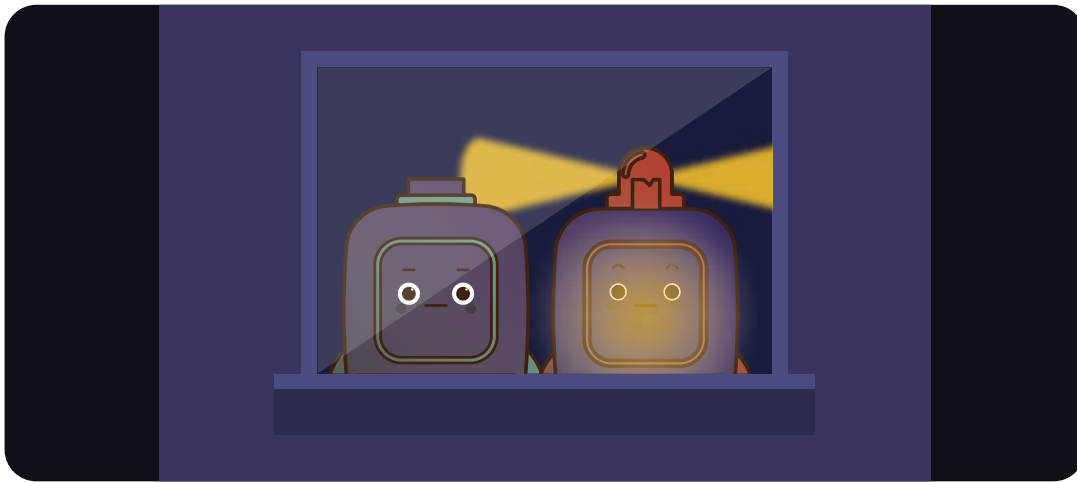
3) Checking the Display: Your Display module will show 4 different types of information. The information is being transmitted from the Environment module to the Display module. Identify each of the information (among temperature, humidity, illuminance, and sound level), and fill in the blanks below.



Let's practice



Let's make a flashlight. Also, identify and categorize each module as either input, output, or setup.

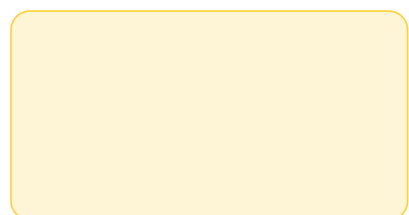
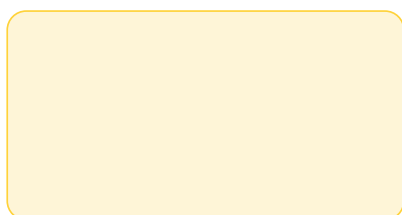
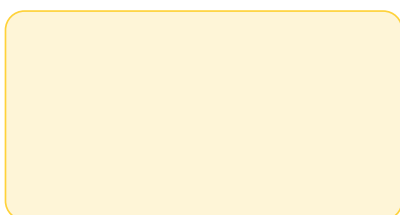
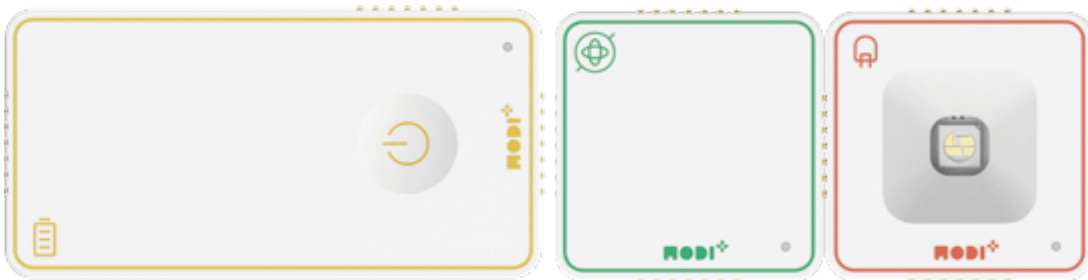
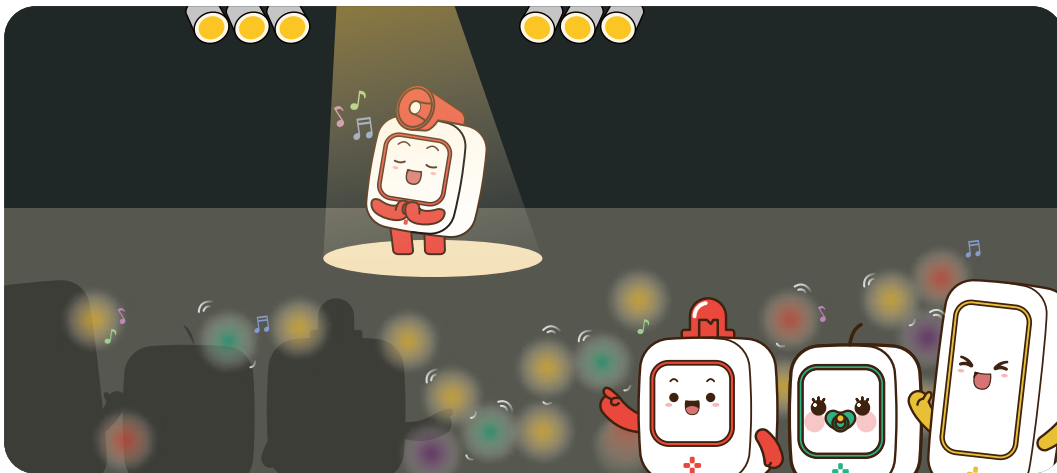




Let's practice



Let's make a rainbow light. Also, identify and categorize each module as either input, output, or setup.



Let's practice



Let's make a rainbow light. Also, identify and categorize each module as either input, output, or setup.

